Speech Recognition

Exercises about Textprocessing Algorithms

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Exercise 01

Implement the Ngram- and Jaccard- algorithms from http://textmining.wp.hs-hannover.de/UnscharfeSuche.html

Combining both, what is the similarity of the words:

"head" and "heat"

"heat" and "hunt"

Use the symbol # for word beginning and $ for ending.

Find more word-pairs that are similar in the sense of this algortihm.

Reduce the number of characters in the word to 2 (bigrams). What are the results now?

Exercise 02

On the same website from exercise 01 there is an implementation of a visualization of the levenshtein-table.

Implement it and apply it for the words "Hallo" and "House".

Use the visualization to understand the algorithm.

In the original version the cost for a substitution is 1. What happens if you increase the substitution cost to 2?

Exercise 03

Using NLTK, compare Porters stemming with cistem stemming on english and german words.

english words: goes, wanted, housekeeping

german: gehen, ging, gewünscht, haushälterin

Exercise 04

Implement the byte-pair encoding using the article at https://www.aclweb.org/anthology/P16-1162/

Debug and step through the code to see how the algorithm works.

Practical

Part 01

Extend your chatbot so that it outputs the stems and lemmas of the nouns of users' input.